Toward Scalable Bayesian Uncertainty for Machine Learning with the Ray Tracing Sampler

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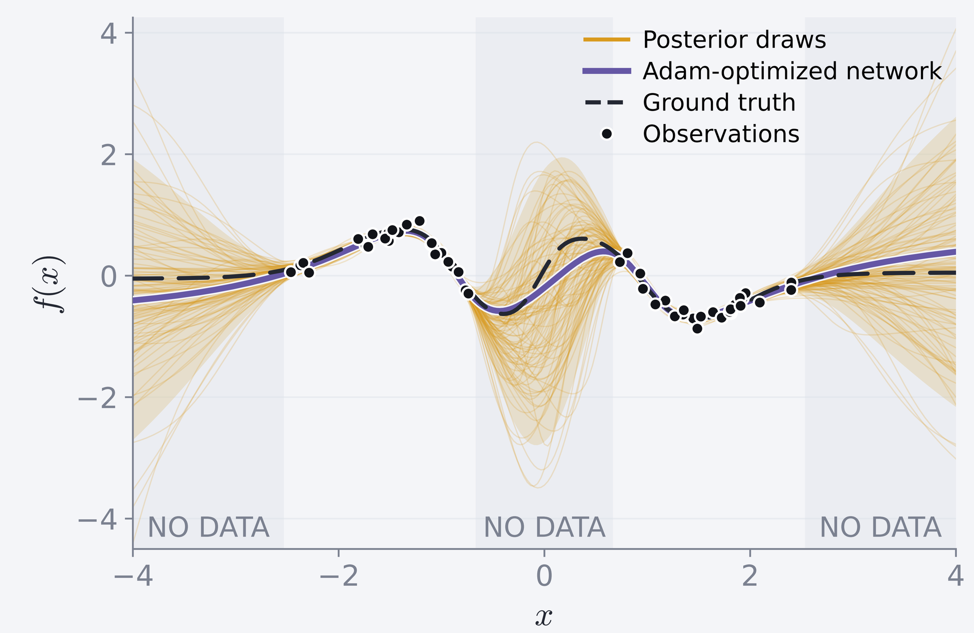
Why sample a neural network?

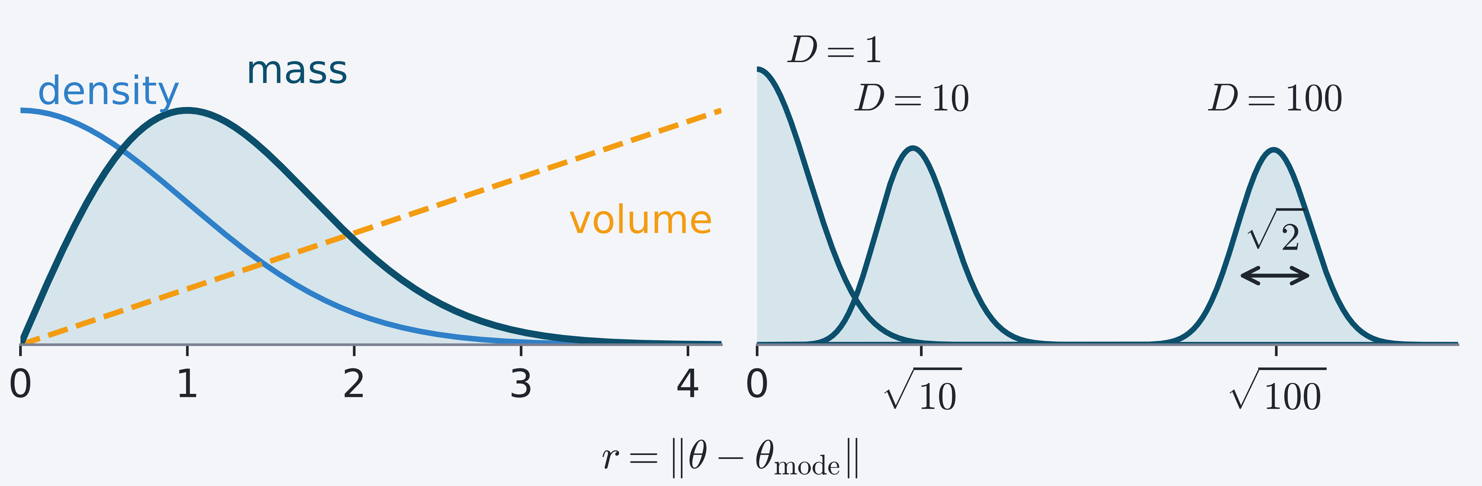


Ordinary training: one set of weights → one function.
Bayesian sampling: a distribution over weights → uncertainty over functions.

$$\pi(w) \;=\; p(w \mid \mathcal{D}) \;\propto\; \underbrace{p(\mathcal{D} \mid w)}_{\text{fit}}\; \underbrace{p(w)}_{\text{prior}}$$

Where the probability lives

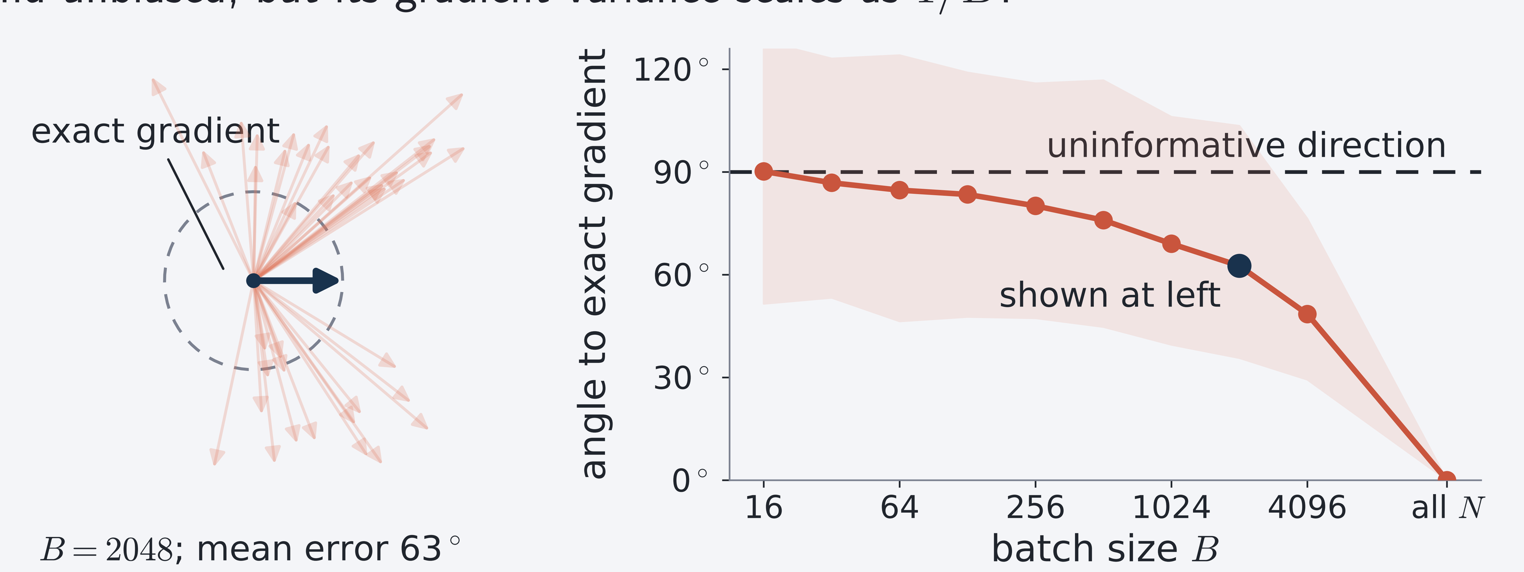
$D = 2$: density $\times$ volume $=$ mass   Higher D: mass peaks at $r \approx \sqrt{D}$



Because volume grows as r^{D-1}, probability concentrates in the *typical set*: a thin shell near $\|\theta\|^2 \approx D$. A correct sampler therefore spends almost all its time on this shell.

MCMC in the era of big data

MCMC generates a sequence of correlated samples targeting π. Exact gradients require all N data points. A minibatch of size $B \ll N$ is N/B times cheaper and unbiased, but its gradient variance scales as $1/B$.



$B = 2048$; mean error 63°

At small B, the mean angular error approaches 90°, so the estimate provides almost no information about the exact gradient direction.

HMC and ray tracing under gradient noise

HMC[1] and ray tracing[2] both use $\nabla \ln \pi$. HMC updates an unconstrained velocity; ray tracing rotates a unit direction.

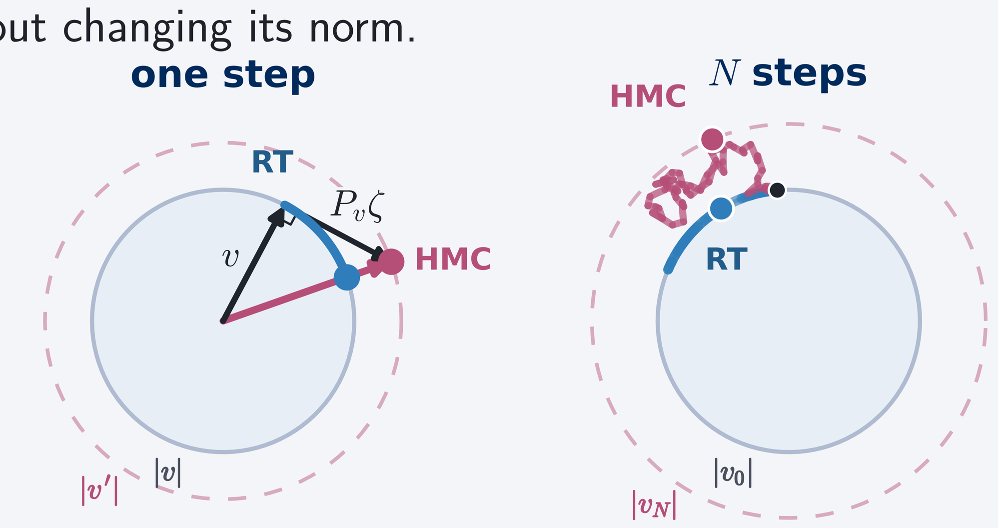
1. One noisy step

Mean-zero gradient error raises HMC's expected kinetic energy, heating the chain and pushing it *outward* off the shell.

For ray tracing, $n(\theta) = \pi(\theta)^{1/(D-1)}$ and the unit direction obeys

$$\frac{\mathrm{d}u}{\mathrm{d}s} = P_u\left(\nabla \ln n(\theta) + \frac{\sigma}{D-1}\zeta\right), \qquad P_u = I - uu^\top.$$

Since P_u removes the component along u, gradient noise rotates the direction without changing its norm.



Live demo

2. Accumulation over many steps

With $b(h, \sigma)$ the equilibrium bias and $\sigma_c(h) \sim h^{-p}$ the noise tolerated at step h: each kick contributes $\sigma^2 h^2 / 2(D-1)^2$ of diffusion Δ_S over directions, and a trajectory of length S holds S/h of them:

$$\frac{S}{h} \cdot \frac{\sigma^2 h^2}{2(D-1)^2} \Delta_S \;=\; O(\sigma^2 h)\,\Delta_S,$$

the order in h that leaves HMC at $p = \frac{1}{2}$.

3. Effect on the stationary distribution

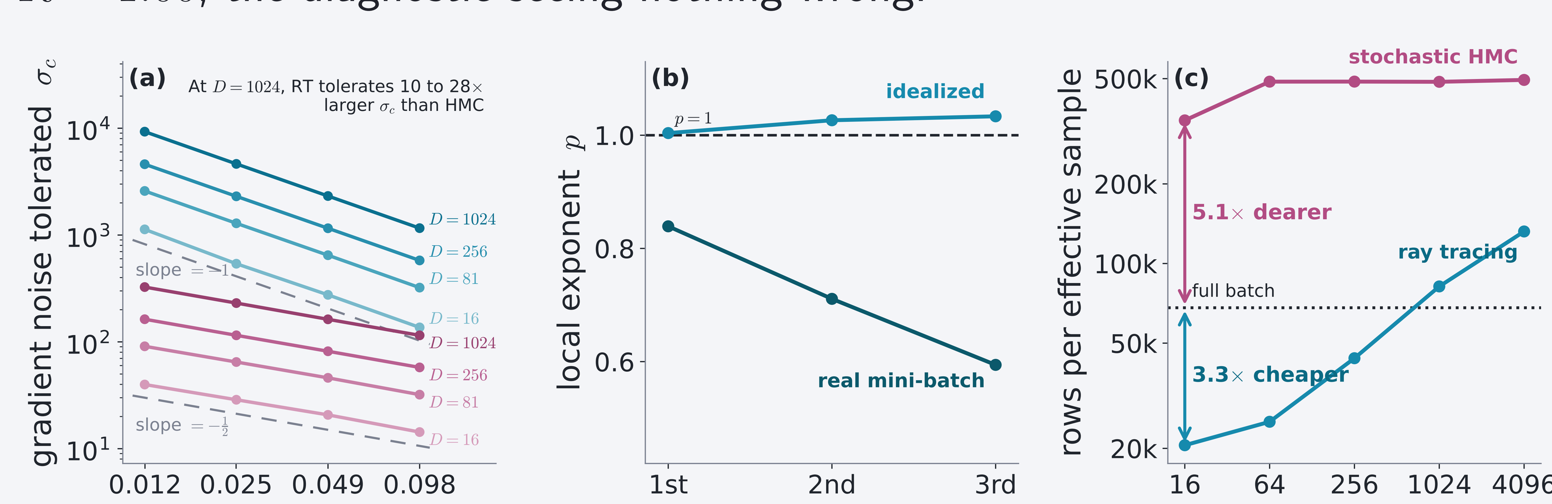
$$\begin{aligned} \rho_0(\theta, u) &= \pi(\theta)\,\mu_S(u), \qquad \Delta_S \rho_0 = 0 \\ \implies\ & b(h, \sigma) = O(\sigma^2 h^2) + o(\sigma^2) \\ \implies\ & \sigma_c(h) = \Omega(h^{-1}), \quad p \geq 1 \end{aligned}$$

Diffusion cannot spread directions that are already uniform.

Individual paths wander at $O(\sigma^2 h)$, but the distribution they settle into does not move at that order.

Measured: $p = 1.00$; HMC at $p = \frac{1}{2}$

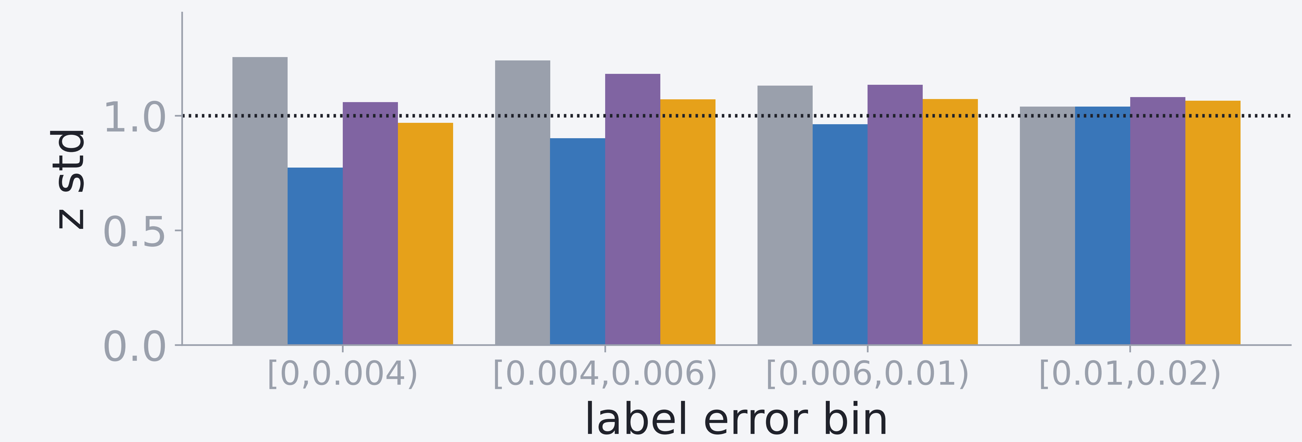
Fitted at every D from 16 to 1024: $p = 1.00$, against $p = \frac{1}{2}$ for HMC. At $\sigma = 20$, stochastic-gradient HMC also settles three times off the shell at split $\hat{R} = 1.00$, the diagnostic seeing nothing wrong.

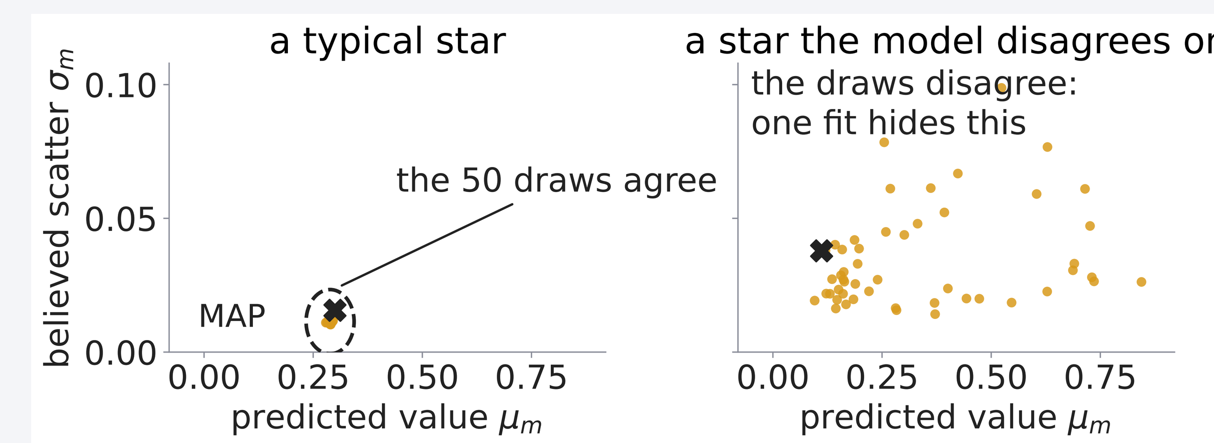


(a) Gaussian target. (b,c) UCI superconductivity.

Uncertainty quality on the Gaia posterior

Result: only ray tracing keeps the error bars right in every group of stars, and averaging its 50 draws makes each held-out star 27% more probable than the single best-fit network (MAP). Gaia XP regression[4], 126,156 stars, $D = 11{,}394$, sampled with minibatch ray tracing. Key: point (MAP), deep ensemble, SGHMC[3], ray tracing.





Conclusion

- Gradient noise can only rotate a unit ray, and the fitted step exponent confirms the bound: $p = 1$ at every dimension tested ($D = 16$ to 1024), against $p = \frac{1}{2}$ for stochastic HMC.
- Real minibatches break the noise assumptions and erode the clean exponent, yet the compute advantage survives ($3.3\times$ cheaper per effective sample; stochastic HMC $5.1\times$ dearer). Still open: whether $p = 1$ is exact in stationarity.
- On a real survey posterior, only ray tracing keeps per-group calibration flat, and averaging its draws makes each held-out star 27% more probable than the single best fit.

References

[1] Betancourt, M. 2017, *A Conceptual Introduction to Hamiltonian Monte Carlo*, arXiv:1701.02434

[2] Behroozi, P. 2026, *The Ray Tracing Sampler: Bayesian Sampling of Neural Networks for Everyone*, Open Journal of Astrophysics 9, doi:10.33232/001c.165932

[3] Chen, T., Fox, E. B., Guestrin, C. 2014, *Stochastic Gradient Hamiltonian Monte Carlo*, ICML, PMLR 32(2), 1683

[4] Gaia Collaboration, Vallenari, A., et al. 2023, *Gaia Data Release 3: Summary of the content and survey properties*, A&A 674, A1

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